

Image Generation, Layout Builder & PDF Export Module

5.5 Image Generation Module

The Image Generation Module creates comic-style illustrations using Stable Diffusion. Scene descriptions generated by the Story Generation Module are converted into high-quality comic illustrations for each panel. Images are stored temporarily and linked with their respective panels for layout generation.

Responsibilities

- Generate comic-style illustrations.
- Process scene descriptions.
- Convert prompts into AI-generated images.
- Store generated illustrations.
- Return image paths to the backend.
- Maintain visual consistency.

5.6 Layout Builder Module

The Layout Builder combines narration, dialogues, titles, and illustrations into sequential comic panels. It prepares the comic preview and ensures consistent formatting before PDF generation.

Responsibilities

- Combine text and images.
- Organize comic panels.
- Prepare preview layout.
- Maintain panel sequence.
- Prepare final layout for PDF generation.

5.7 PDF Export Module

Using the FPDF library, the PDF Export Module converts the completed comic into a professionally formatted PDF suitable for downloading, printing, and sharing.

Responsibilities

- Generate PDF.
- Arrange comic panels.
- Insert illustrations.
- Save exported document.
- Return download path.

Module Integration Summary

These modules work together to transform AI-generated stories and illustrations into a complete downloadable comic.